

Counting terms in derivatives of x^x

An exact reduction of OEIS A293239, nonvanishing theorems,
computational certificates, and a corrected recurrence

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Abstract

The sequence A293239 counts the nonzero terms in the fully expanded n th derivative of x^x . Its proposed generating function is rational, and its proposed eventual formula is a quadratic quasipolynomial of period four. We reduce the counting problem exactly to the zero set of the first-kind Lehmer–Comtet numbers. All cancellations predicted by the conjecture are proved: a symmetry family and one additional zero. The converse assertion, excluding every further zero, is not established here; consequently this report does *not* claim a complete proof of the principal closed formula. The unconditional results include an exact defect formula, upper and lower bounds proving quadratic-order growth, nonvanishing on every prime-offset diagonal with an exact prime-adic valuation, and complete zero classifications for the first three columns and offsets one through sixteen. Independently implemented exact computations confirm the formula through $n = 3000$. A separate conjectured recurrence in the OEIS entry is false with its printed range: it already fails at $n = 8$. For the proposed sequence its correct eventual starting point is $n = 14$. The accompanying code and certificates distinguish finite verification from statements proved for infinitely many indices.

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1 The question and the status of the answer

For $x > 0$, put $f(x) = x^x = \exp(x \log x)$. Let $a(n)$ be the number of nonzero summands after $f^{(n)}(x)$ is expanded into the monomials

$$x^{x-k}(\log x)^j, \quad k, j \in \mathbb{Z}_{\geq 0}, \quad (1.1)$$

and coefficients of identical monomials are collected. In particular, an unexpanded expression such as $x^x(1 + \log x)^n$ is not one term for this purpose. This is the convention in OEIS A293239, introduced by Vladimir Reshetnikov in 2017 [1].

The entry attributes the rational generating function and closed formula to Colin Barker and the proposed asymptotic equivalent to Václav Kotešovec. To separate a proved object from the proposed answer, define

$$q(0) = 1, \quad q(n) = 1 + \frac{n(n+1)}{2} - \left\lfloor \frac{n-1}{4} \right\rfloor - \mathbf{1}_{\{n \geq 8\}} \quad (n \geq 1). \quad (1.2)$$

The principal conjecture is $a(n) = q(n)$. For $n \geq 8$, this says simply

$$a(n) \stackrel{?}{=} \frac{n(n+1)}{2} - \left\lfloor \frac{n-1}{4} \right\rfloor. \quad (1.3)$$

The proposed generating function is

$$Q(z) := \sum_{n \geq 0} q(n)z^n = \frac{1 + z^2 + z^3 + z^6 - z^8 + z^9 + z^{12} - z^{13}}{(1-z)^2(1-z^4)}. \quad (1.4)$$

The rational function in (1.4) is indeed the generating function of the explicitly defined sequence q . What remains to be proved is its identification with the derivative count a .

1.1 What is proved

The central unconditional identity will be

$$a(n) = q(n) - E(n), \quad (1.5)$$

where $E(n)$ counts a precisely defined set of additional zero coefficients. In particular, $E(n)$ is a nonnegative, nondecreasing integer. Thus the proposed answer is an upper bound, not a formula that can be exceeded by a previously overlooked cancellation.

We prove, for all $n \geq 0$,

$$n + 1 + \left\lfloor \frac{n^2}{4} \right\rfloor \leq a(n) \leq q(n). \quad (1.6)$$

Consequently $a(n) = \Theta(n^2)$. This does not establish the sharper claim $a(n) \sim n^2/2$; that claim is equivalent to $E(n) = o(n^2)$. We also prove infinite nonvanishing results that substantially restrict where additional zeros could lie. The remaining gap is stated explicitly in Conjecture 4.4.

1.2 A correction that is unconditional

The OEIS entry also proposes

$$a(n) = 2a(n-1) - a(n-2) + a(n-4) - 2a(n-5) + a(n-6) \quad (n > 6). \quad (1.7)$$

This range is incorrect. Exact low-order differentiation gives

$$a(2) = 4, \quad a(3) = 7, \quad a(4) = 11, \quad a(6) = 21, \quad a(7) = 28, \quad a(8) = 35,$$

whereas the right side of (1.7) at $n = 8$ is

$$2 \cdot 28 - 21 + 11 - 2 \cdot 7 + 4 = 36 \neq 35.$$

For q , the recurrence holds for every $n \geq 14$, and it fails at $n = 13$. This correction does not disprove the proposed generating function or closed formula. Section 8 explains the distinction.

2 A canonical polynomial for each derivative

Set $t = x^{-1}$ and $L = \log x$. There is a polynomial $P_n \in \mathbb{Z}[t, L]$ such that

$$f^{(n)}(x) = x^x P_n(x^{-1}, \log x). \quad (2.1)$$

Differentiating this expression and using $t' = -t^2$, $L' = t$, and $(x^x)' = x^x(1 + L)$ gives the following exact recurrence.

Proposition 2.1. *The derivative polynomials satisfy*

$$P_0 = 1, \quad P_{n+1} = (1 + L)P_n + t \frac{\partial P_n}{\partial L} - t^2 \frac{\partial P_n}{\partial t}. \quad (2.2)$$

In particular, they have integer coefficients and can be computed using integer arithmetic only.

For example,

$$\begin{aligned} P_1 &= 1 + L, \\ P_2 &= 1 + 2L + L^2 + t, \\ P_3 &= 1 + 3L + 3L^2 + L^3 + 3t + 3tL - t^2. \end{aligned}$$

Thus $a(3) = 7$, consistently with the example in the entry.

Lemma 2.2. *The monomials in (1.1) are linearly independent over $\mathbb{R}$ when only finitely many are used. Consequently the collected expansion is unique, and $a(n) = |\text{supp } P_n|$.*

Proof. Divide a putative relation by the positive function x^x and set $x = e^s$. The result is $\sum_{k=0}^K e^{-ks} p_k(s) = 0$, with polynomial p_k . If some p_k is nonzero, let k_0 be the smallest such index. Multiplication by $e^{k_0 s}$ shows that $p_{k_0}(s)$ tends to zero as $s \rightarrow +\infty$, because all remaining summands decay exponentially times a polynomial. No nonzero polynomial has that property. This is a contradiction. $\square$

The independence statement matters: the variables t and L are related by $t = e^{-L}$ after substitution, but this relation causes no additional finite polynomial identities.

3 The Lehmer–Comtet coefficient triangle

3.1 Definition and generating function

Put

$$\Phi(z) = (1 + z) \log(1 + z) = z + \sum_{d \geq 1} \frac{(-1)^{d-1}}{d(d+1)} z^{d+1}. \quad (3.1)$$

Define

$$b(m, r) = \frac{m!}{r!} [z^m] \Phi(z)^r \quad (m, r \geq 0). \quad (3.2)$$

Here $b(0, 0) = 1$, $b(m, 0) = 0$ for $m > 0$, and $b(m, r) = 0$ when $r > m$. These are the first-kind Lehmer–Comtet numbers, tabulated in OEIS A008296 [2]. The general derivative problem and these numbers belong to the classical literature of Comtet, Lehmer, and Gould [3, 4, 5]. A recent treatment of their identities is Jeong’s preprint [6]. We derive the identities needed here rather than assuming a nonvanishing conclusion from those formulas.

Theorem 3.1 (Exact coefficient formula). *For every $n \geq 0$,*

$$f^{(n)}(x) = x^x \sum_{m=0}^n \binom{n}{m} (\log x)^{n-m} \sum_{r=0}^m b(m, r) x^{r-m}. \quad (3.3)$$

Equivalently,

$$[t^k][L^j]P_n(t, L) = \binom{n}{j} b(n-j, n-j-k), \quad (3.4)$$

with out-of-range entries interpreted as zero.

Proof. Taylor expansion at a fixed $x > 0$ gives

$$\sum_{n \geq 0} \frac{f^{(n)}(x)}{x^x} \frac{u^n}{n!} = \frac{(x+u)^{x+u}}{x^x} = \exp(u \log x) \exp(x\Phi(u/x)).$$

Using (3.2), expand the second exponential:

$$\begin{aligned} \exp(x\Phi(u/x)) &= \sum_{r \geq 0} \frac{x^r}{r!} \Phi(u/x)^r \\ &= \sum_{m \geq 0} \frac{u^m}{m!} \sum_{r=0}^m b(m, r) x^{r-m}. \end{aligned}$$

Multiplication by $\exp(u \log x)$ and extraction of the coefficient of u^n proves (3.3). All manipulations can also be performed formally, since only finitely many terms affect any prescribed coefficient. $\square$

3.2 A shifted falling-factorial formula

Let

$$F_m(y) = y^{\overline{m}} = \prod_{h=0}^{m-1} (y-h), \quad F_0(y) = 1.$$

Proposition 3.2. *For $0 \leq r \leq m$,*

$$b(m, r) = \frac{F_m^{(r)}(r)}{r!} = [u^r] \prod_{h=0}^{m-1} (u+r-h). \quad (3.5)$$

In particular, all $b(m, r)$ are integers.

Proof. The binomial-series identity is $(1+z)^y = \sum_{m \geq 0} F_m(y) z^m / m!$. Differentiate r times in y and set $y = r$. The left side becomes $(1+z)^r \log(1+z)^r = \Phi(z)^r$. Comparing coefficients gives the first formula. The second is the polynomial Taylor formula at $y = r$. $\square$

Writing $F_m(y) = \sum_{\ell=0}^m s(m, \ell) y^\ell$ in terms of the signed Stirling numbers of the first kind also gives

$$b(m, r) = \sum_{\ell=r}^m \binom{\ell}{r} r^{\ell-r} s(m, \ell). \quad (3.6)$$

This is another exact formula, but its alternating summands do not by themselves resolve whether the result can vanish.

3.3 An integer recurrence

Proposition 3.3. For $m \geq 2$ and $1 \leq r \leq m$,

$$b(m, r) = (r - m + 1)b(m - 1, r) + b(m - 1, r - 1) + (m - 1)b(m - 2, r - 1). \quad (3.7)$$

Together with rows $b(0, 0) = 1$ and $(b(1, 0), b(1, 1)) = (0, 1)$, this determines the entire triangle.

Proof. Let $B_r(z) = \Phi(z)^r/r!$. Since $(1 + z)\Phi'(z) = 1 + z + \Phi(z)$, we have

$$(1 + z)B'_r(z) = (1 + z)B_{r-1}(z) + rB_r(z).$$

Compare coefficients of $z^{m-1}/(m-1)!$. The result is

$$b(m, r) + (m - 1)b(m - 1, r) = b(m - 1, r - 1) + (m - 1)b(m - 2, r - 1) + rb(m - 1, r),$$

which rearranges to (3.7). $\square$

For reference, the first rows, omitting column zero, are

$m \backslash r$	1	2	3	4	5	6	7	8	9
1	1								
2	1	1							
3	-1	3	1						
4	2	-1	6	1					
5	-6	0	5	10	1				
6	24	4	-15	25	15	1			
7	-120	-28	49	-35	70	21	1		
8	720	188	-196	49	0	154	28	1	
9	-5040	-1368	944	0	-231	252	294	36	1

4 The exact zero-count reduction

Definition 4.1. For $n \geq 0$, let

$$Z(n) = \#\{(m, r) : 1 \leq r \leq m \leq n, b(m, r) = 0\}.$$

Column zero is not included: its zeros for $m > 0$ are forced by the definition and never represent potential nonzero derivative terms.

Theorem 4.2. For every $n \geq 0$,

$$a(n) = 1 + \frac{n(n+1)}{2} - Z(n). \quad (4.1)$$

Proof. In (3.3), the pair (m, r) gives the monomial

$$x^{x-(m-r)}(\log x)^{n-m}$$

with coefficient $\binom{n}{m}b(m, r)$. The map $(m, r) \mapsto (m - r, n - m)$ is injective for fixed n : the second coordinate recovers m , and the first then recovers r . Thus different pairs do not cancel each other. The factor $\binom{n}{m}$ is positive. There is one nonzero entry at $(0, 0)$, and m potentially nonzero entries $1 \leq r \leq m$ in every row $m \geq 1$. Removing exactly the zero entries proves the formula. $\square$

This argument isolates the main difficulty. Once the triangle is known, term counting is easy; proving that its list of zeros is complete is not a consequence of merely writing down the triangle's recurrence.

4.1 Cancellations that are proved

Proposition 4.3 (Symmetry zeros). *For every integer $h \geq 1$,*

$$b(4h + 1, 2h) = 0. \quad (4.2)$$

Proof. Center the product in (3.5) at $r = 2h$:

$$F_{4h+1}(2h + u) = u \prod_{s=1}^{2h} (u^2 - s^2).$$

This polynomial is odd. Its coefficient of the even power u^{2h} is zero. $\square$

This family is not claimed as new: it appears as Proposition 3.10(iii) in Jeong [6], with attribution there to Theorem 7 of Lehmer [4]. Its elementary proof is included to make the reduction self-contained.

There is one further zero required by the proposed formula:

$$b(8, 5) = 0. \quad (4.3)$$

It follows immediately from the factorization proved in Section 5:

$$b(m, m - 3) = \frac{m(m - 1)(m - 2)(m - 3)(m - 5)(m - 8)}{48}.$$

The root $m = 5$ is the first symmetry zero; $m = 8$ is genuinely additional.

4.2 The precise missing assertion

Conjecture 4.4 (Complete zero classification). *For integers $1 \leq r \leq m$,*

$$b(m, r) = 0 \iff (m, r) = (8, 5) \text{ or } (m, r) = (4h + 1, 2h) \text{ for some } h \geq 1. \quad (4.4)$$

The right-to-left implication is proved above. The left-to-right implication is the unproved step in this report.

Let

$$\mathcal{Z}_0 = \{(8, 5)\} \cup \{(4h + 1, 2h) : h \geq 1\}$$

be the proved set of zeros, and define

$$e_m = \#\{r : 1 \leq r \leq m, b(m, r) = 0, (m, r) \notin \mathcal{Z}_0\},$$

$$E(n) = \sum_{m=1}^n e_m.$$

Thus E counts only zeros not in the proved list.

Theorem 4.5 (Exact defect formula). *For every $n \geq 0$,*

$$a(n) = q(n) - E(n).$$

The following are equivalent: Conjecture 4.4; $E(n) = 0$ for all n ; $a(n) = q(n)$ for all n ; and the proposed eventual formula (1.3) for every $n \geq 8$.

Proof. For $n \geq 1$, the number of symmetry zeros with first coordinate at most n is $\lfloor (n-1)/4 \rfloor$. The zero $(8, 5)$ contributes $\mathbf{1}_{\{n \geq 8\}}$. Substituting

$$Z(n) = \left\lfloor \frac{n-1}{4} \right\rfloor + \mathbf{1}_{\{n \geq 8\}} + E(n)$$

into (4.1) proves the identity for $n \geq 1$; $n = 0$ is immediate. The first three equivalences follow from the definitions. The eventual formula also forces $E(n) = 0$ for every $n \geq 8$, and nonnegativity and monotonicity then force $E(n) = 0$ for smaller n as well. $\square$

A hypothetical new zero at row M would decrease every count $a(n)$ for $n \geq M$ relative to $q(n)$. It would not merely alter a single isolated term of the sequence. This persistent-deficit property is a useful safeguard when testing proposed proofs or numerical data.

5 Diagonal polynomials and exact nonvanishing

5.1 A polynomial for each fixed offset

Define

$$H(z) = \frac{\Phi(z)}{z} = 1 + \frac{z}{2} - \frac{z^2}{6} + \frac{z^3}{12} - \frac{z^4}{20} + \cdots, \quad T_d(Y) = [z^d]H(z)^Y. \quad (5.1)$$

For a fixed integer $d \geq 0$, T_d is a polynomial over $\mathbb{Q}$ in Y .

Lemma 5.1. For $m \geq d$,

$$b(m, m-d) = m^d T_d(m-d). \quad (5.2)$$

For $d \geq 1$, $T_d(Y)$ has degree d , leading coefficient $1/(2^d d!)$, and a factor Y . Consequently $T_d(Y)/Y$ is a nonzero polynomial of degree $d-1$.

Proof. Since $\Phi(z)^{m-d} = z^{m-d}H(z)^{m-d}$, definition (3.2) gives (5.2). Now assume $d \geq 1$ and write $h(z) = H(z) - 1$. Then

$$T_d(Y) = \sum_{\ell=1}^d \binom{Y}{\ell} [z^d]h(z)^\ell. \quad (5.3)$$

Every summand is divisible by Y . The only term of degree d in Y comes from $\ell = d$, and $[z^d]h(z)^d = 2^{-d}$. This proves the remaining assertions. $\square$

The first three cases are

$$b(m, m-1) = \frac{m(m-1)}{2}, \quad (5.4)$$

$$b(m, m-2) = \frac{m(m-1)(m-2)(3m-13)}{24}, \quad (5.5)$$

$$b(m, m-3) = \frac{m(m-1)(m-2)(m-3)(m-5)(m-8)}{48}. \quad (5.6)$$

For instance,

$$T_3(Y) = \frac{Y}{12} - \frac{Y(Y-1)}{12} + \frac{Y(Y-1)(Y-2)}{48} = \frac{Y(Y-2)(Y-5)}{48}.$$

This proves both zeros on the third offset and excludes all others on that entire diagonal, not just up to a numerical cutoff.

5.2 Prime offsets: an infinite family of nonvanishing results

For a nonzero rational number v , write $v_p(v)$ for its exponent of the prime p in its prime factorization. A rational number is called p -integral when its denominator in lowest terms is not divisible by p .

Theorem 5.2 (Prime-offset nonvanishing). *Let p be any prime and let $m \geq p$. Then*

$$b(m, m - p + 1) \neq 0, \quad v_p(b(m, m - p + 1)) = v_p\left(\left\lfloor \frac{m}{p} \right\rfloor\right). \quad (5.7)$$

In particular, an additional zero cannot occur on any offset $d = m - r$ for which $d + 1$ is prime.

Proof. Put $d = p - 1$ and $h_j = [z^j](H(z) - 1)$. For $1 \leq j \leq p - 2$, the number $h_j = (-1)^{j-1}/(j(j+1))$ is p -integral. Moreover,

$$ph_{p-1} \equiv 1 \pmod{p}.$$

For odd p , this follows from $ph_{p-1} = -1/(p-1)$; for $p = 2$, it is $2h_1 = 1$.

Divide (5.3) by Y . Its $\ell = 1$ summand is h_{p-1} . For every $2 \leq \ell \leq p - 1$, a composition of $p - 1$ into ℓ positive parts has all parts at most $p - 2$. Therefore every coefficient contributing to $[z^{p-1}]h(z)^\ell$ is p -integral. Also,

$$\frac{1}{Y} \binom{Y}{\ell} = \frac{(Y-1) \cdots (Y-\ell+1)}{\ell!}$$

has p -integral coefficients, because $\ell!$ is not divisible by p . It follows, as a polynomial congruence over the p -integral rationals, that

$$p \frac{T_{p-1}(Y)}{Y} \equiv 1 \pmod{p}. \quad (5.8)$$

For every positive integer Y , the left side is therefore nonzero and has p -adic valuation zero. Hence $v_p(T_{p-1}(Y)) = v_p(Y) - 1$.

Set $Y = m - p + 1 > 0$ and use (5.2). This yields

$$v_p(b(m, m - p + 1)) = v_p(m^{\underline{p}}) - 1.$$

Among the p consecutive positive integers $m - p + 1, \dots, m$, exactly one is divisible by p : it is $p \lfloor m/p \rfloor$. Its valuation is $1 + v_p(\lfloor m/p \rfloor)$, proving (5.7). $\square$

This proof explains, for example, why the residual polynomial $3m - 13$ in (5.5) is a nonzero constant modulo 3, and why analogous residual polynomials for offsets 4, 6, 10, 12, 16 are nonzero constants modulo 5, 7, 11, 13, 17, respectively.

5.3 Finite certificates that cover infinitely many row indices

For a fixed $d \geq 1$, put

$$S_d(X) = T_d(X - d).$$

The factor $X - d$ is forced by Lemma 5.1. When $d \geq 3$ is odd, the symmetry zero also forces $X - (2d - 1)$; for $d = 3$ there is in addition the factor $X - 8$. Remove these factors from S_d , clear rational denominators, divide out the greatest common divisor of the integer coefficients, and normalize the leading coefficient to be positive. Call the resulting polynomial $R_d(X)$. Thus, with a positive rational constant c_d ,

$$S_d(X) = c_d(X - d) \begin{cases} (X - 5)(X - 8)R_d(X), & d = 3, \\ (X - (2d - 1))R_d(X), & d \geq 5 \text{ odd}, \\ R_d(X), & \text{otherwise.} \end{cases} \quad (5.9)$$

Lemma 5.3 (Modular certificate). *If $R \in \mathbb{Z}[X]$ satisfies $R(t) \not\equiv 0 \pmod{p}$ for every $t \in \{0, 1, \dots, p-1\}$, then R has no integer root.*

Proof. An integer root, reduced modulo p , would be one of those residues and would give a zero value modulo p . $\square$

Theorem 5.4 (Computer-assisted diagonal classification). *Conjecture 4.4 holds for every pair $1 \leq r \leq m$ with $m - r \leq 16$.*

Proof. For offset zero, $b(m, m) = 1$. For $1 \leq d \leq 16$, compute S_d exactly from (5.3) and factor it as in (5.9). The supplied `data/diagonal_certificates.json` records the full rational coefficients of S_d , the scalar c_d , all removed linear factors, the full integer coefficients of R_d , and every value of $R_d(t)$ modulo the certifying prime. All of those values are nonzero. The following summary identifies the certificates:

d	$\deg R_d$	p	roots mod p	d	$\deg R_d$	p	roots mod p
1	0	2	0	9	7	7	0
2	1	3	0	10	9	11	0
3	0	2	0	11	9	13	0
4	3	5	0	12	11	13	0
5	3	3	0	13	11	11	0
6	5	7	0	14	13	23	0
7	5	5	0	15	13	19	0
8	7	11	0	16	15	17	0

The standard-library program `verify_diagonals.py` reconstructs every polynomial using rational arithmetic and checks the factorization and each residue. The modular lemma excludes every integer root of every R_d . For $m > d$, the removed factor $m - d$ cannot vanish; the remaining removed factors give exactly the stated symmetry and exceptional zeros. $\square$

These certificates are not extrapolations from finitely many values of m : each modular argument excludes integer roots for *all* m on its fixed diagonal. What is finite is the set of offsets treated.

For a small example not requiring the machine-readable file,

$$\begin{aligned} R_4(X) &= 15X^3 - 390X^2 + 3245X - 8638 \equiv 2 \pmod{5}, \\ R_5(X) &= 3X^3 - 103X^2 + 1118X - 3776 \equiv 2X^2 + 2X + 1 \pmod{3}. \end{aligned}$$

The second reduced polynomial takes the values 1, 2, 1 at the three residues modulo 3. Hence there are no additional zeros on offset 4, and the only zero on offset 5 is the symmetry zero $m = 9$.

6 The first three columns for all row indices

The diagonal results keep $m - r$ fixed. A complementary calculation keeps r fixed and proves nonvanishing in a different infinite region.

Proposition 6.1. *Column $r = 1$ has no zero. Column $r = 2$ has exactly the zero $m = 5$. Column $r = 3$ has no zero.*

Proof. For $r = 1$, (3.5) gives

$$b(1, 1) = 1, \quad b(m, 1) = (-1)^m(m-2)! \quad (m \geq 2).$$

For $r = 2$, put $t = m - 3 \geq 0$. The shifted product is

$$F_m(2+u) = u(u+1)(u+2) \prod_{j=1}^t (u-j).$$

The coefficient of u^2 is

$$b(m, 2) = (-1)^m (m-3)! (2H_{m-3} - 3), \quad m \geq 3, \quad (6.1)$$

where $H_t = \sum_{j=1}^t 1/j$ and $H_0 = 0$. The harmonic numbers strictly increase, and $H_2 = 3/2$, so the only zero is $m - 3 = 2$. The remaining boundary entry is $b(2, 2) = 1$.

For $r = 3$, the boundary entry is $b(3, 3) = 1$. Put $t = m - 4 \geq 0$ and define

$$H_t^{(2)} = \sum_{j=1}^t \frac{1}{j^2}, \quad g(t) = 6 - 11H_t + 3(H_t^2 - H_t^{(2)}).$$

Extracting the coefficient of u^3 from $u(u+1)(u+2)(u+3) \prod_{j=1}^t (u-j)$ gives

$$b(m, 3) = (-1)^t t! g(t). \quad (6.2)$$

The elementary identity

$$g(t+1) - g(t) = \frac{6H_t - 11}{t+1} \quad (6.3)$$

controls its entire sign pattern. We have $g(0) = 6$, $g(1) = -5$, and $H_3 = 11/6$. Thus g decreases through $t = 3$, has $g(3) = g(4) = -49/6$, and is strictly increasing thereafter. Finally, exact rational evaluation gives

$$g(19) = -\frac{4571462267}{97772875200} < 0, \quad g(20) = \frac{537826687}{1150269120} > 0.$$

It follows that $g(t) < 0$ for $1 \leq t \leq 19$, whereas $g(t) > 0$ for $t \geq 20$. There is no integer zero. $\square$

The last argument illustrates a distinction from a numerical search: exact values at two adjacent integers, together with a proved monotonicity statement outside a finite initial segment, settle the entire column.

7 Unconditional quadratic growth

Theorem 7.1. *For every integer $n \geq 0$,*

$$n + 1 + \left\lfloor \frac{n^2}{4} \right\rfloor \leq a(n) \leq q(n). \quad (7.1)$$

In particular,

$$\frac{1}{4} \leq \liminf_{n \rightarrow \infty} \frac{a(n)}{n^2} \leq \limsup_{n \rightarrow \infty} \frac{a(n)}{n^2} \leq \frac{1}{2}, \quad a(n) = \Theta(n^2).$$

Proof. The upper bound is Theorem 4.5. For the lower bound, count nonzero entries along diagonals of the triangle in Theorem 4.2. The main diagonal, including $(0, 0)$, contributes $n + 1$ entries.

Fix $1 \leq d \leq n - 1$. The diagonal $m - r = d$ has $n - d$ relevant positions, $m = d + 1, \dots, n$. By Lemma 5.1, its zero positions are roots of the nonzero polynomial $T_d(m - d)/(m - d)$ of degree $d - 1$. There are therefore at most $d - 1$ such positions, and at least

$$\max(0, n - d - (d - 1)) = \max(0, n - 2d + 1)$$

nonzero entries on the diagonal. Summing over all $d \geq 1$ gives

$$a(n) \geq n + 1 + \sum_{d \geq 1} \max(0, n - 2d + 1) = n + 1 + \left\lfloor \frac{n^2}{4} \right\rfloor.$$

The last equality follows by summing the odd numbers $1, 3, \dots, 2s - 1$ when $n = 2s$, and the even numbers $2, 4, \dots, 2s$ when $n = 2s + 1$. The asymptotic bounds follow immediately. $\square$

Corollary 7.2 (A stronger explicit lower bound). *For every $n \geq 0$,*

$$a(n) \geq n + 1 + \left\lfloor \frac{n^2}{4} \right\rfloor + \sum_{\substack{p \leq n \\ p \text{ prime}}} \min(p - 2, n - p + 1). \quad (7.2)$$

Proof. On offset $d = p - 1$, Theorem 5.2 guarantees all $n - d = n - p + 1$ entries, rather than just $\max(0, n - 2p + 3)$. The improvement is $\min(p - 2, n - p + 1)$. These diagonals are disjoint, so their improvements add. $\square$

The prime-enhanced bound is useful without invoking any theorem about the density of primes. It still does not close the factor-of-two gap in the leading quadratic coefficient. In terms of the exact defect,

$$a(n) \sim \frac{n^2}{2} \iff E(n) = o(n^2). \quad (7.3)$$

Thus the proposed asymptotic equivalent is potentially a weaker target than the complete zero classification: it would allow a sufficiently sparse set of additional zeros.

8 The rational function, quasipolynomial, and recurrence

8.1 Deriving the proposed generating function

There is one potential term at $(0, 0)$ and m potential terms in row m . The cumulative potential count therefore has generating function

$$\sum_{n \geq 0} \left(1 + \frac{n(n+1)}{2}\right) z^n = \frac{1}{1-z} + \frac{z}{(1-z)^3}.$$

The symmetry zeros are born at rows $5, 9, 13, \dots$, and each remains absent from every later derivative. Their cumulative generating function is $z^5/((1-z)(1-z^4))$. The exceptional zero is born at row 8 and contributes $z^8/(1-z)$. Subtracting these two *proved* families gives

$$\begin{aligned} Q(z) &= \frac{1}{1-z} + \frac{z}{(1-z)^3} - \frac{z^5}{(1-z)(1-z^4)} - \frac{z^8}{1-z} \\ &= \frac{1 + z^2 + z^3 + z^6 - z^8 + z^9 + z^{12} - z^{13}}{(1-z)^2(1-z^4)}. \end{aligned} \quad (8.1)$$

This explains every correction term in the proposed rational function.

Theorem 8.1 (Exact generating-function correction). *With $A(z) = \sum_{n \geq 0} a(n)z^n$ and $\mathcal{E}(z) = \sum_{m \geq 1} e_m z^m$,*

$$A(z) = Q(z) - \frac{\mathcal{E}(z)}{1-z}. \quad (8.2)$$

Consequently the proposed rational function equals $A(z)$ if and only if Conjecture 4.4 is true.

Proof. Sum $a(n) = q(n) - \sum_{m \leq n} e_m$ as a formal power series. Each e_m contributes $e_m z^m / (1-z)$ to the cumulative sum. $\square$

8.2 The real form of the complex expression

For $n \geq 8$, division into the four residue classes gives

$$q(n) = \frac{n^2}{2} + \frac{n}{4} + \begin{cases} 1, & n \equiv 0 \pmod{4}, \\ \frac{1}{4}, & n \equiv 1 \pmod{4}, \\ \frac{1}{2}, & n \equiv 2 \pmod{4}, \\ \frac{3}{4}, & n \equiv 3 \pmod{4}. \end{cases} \quad (8.3)$$

The same periodic correction is

$$\frac{5 + (-1)^n + (1-i)(-i)^n + (1+i)i^n}{8}, \quad i^2 = -1.$$

Substitution yields exactly the complex-valued-looking expression printed in OEIS, whose total value is real and integral. This is an algebraic identity for q , independent of the unresolved equality $a = q$.

For all $n \geq 0$, another convenient form is

$$q(n) = \frac{n(n+1)}{2} - \left\lfloor \frac{n-1}{4} \right\rfloor + \mathbf{1}_{\{1 \leq n \leq 7\}}.$$

Here $\lfloor -1/4 \rfloor = -1$ correctly gives $q(0) = 1$. The first terms are

$$1, 2, 4, 7, 11, 15, 21, 28, 35, 43, 53, 64, 76, 88, 102, 117, 133, \dots$$

8.3 Why the recurrence starts at fourteen

Set

$$\begin{aligned} D(z) &= (1-z)^2(1-z^4) = 1 - 2z + z^2 - z^4 + 2z^5 - z^6, \\ N(z) &= 1 + z^2 + z^3 + z^6 - z^8 + z^9 + z^{12} - z^{13}. \end{aligned}$$

The identity $D(z)Q(z) = N(z)$ implies, for $n \geq 6$,

$$q(n) - 2q(n-1) + q(n-2) - q(n-4) + 2q(n-5) - q(n-6) = [z^n]N(z). \quad (8.4)$$

The right side vanishes for every $n \geq 14$, but at $n = 13$ it equals -1 . Thus the least starting index from which the homogeneous recurrence holds for *every* subsequent index of q is 14. For $7 \leq n \leq 13$, its only nonzero residuals are

n	8	9	12	13
residual	-1	1	1	-1

This proves the range correction asserted in Section 1. If the original derivative sequence satisfies the corrected recurrence for all $n \geq 14$, its directly verified initial values $a(0), \dots, a(13)$ force $a = q$ by induction. Consequently the corrected recurrence for a is itself equivalent to the main conjecture, once those initial values are supplied; it is not an independent proved consequence for arbitrary orders.

The second forward differences $q(n+2) - 2q(n+1) + q(n)$ begin

$$1, 1, 1, 0, 2, 1, 0, 1, \underbrace{2, 1, 1, 0, 2, 1, 1, 0, \dots}_{n \geq 8}$$

Again, this is proved for q ; transferring the entire pattern to a requires the missing nonvanishing statement.

9 Reproducible verification

9.1 What was actually checked

All reported checks use exact arithmetic. No floating-point threshold was used to decide that a coefficient was zero.

The C++ implementation with GMP computes the recurrence (3.7) through row 3000. It tests all 4,501,500 entries with $1 \leq r \leq m \leq 3000$. Exactly 750 are zero: 749 symmetry zeros and (8, 5). There are no other zeros in that range, and therefore

$$a(n) = q(n) \quad (0 \leq n \leq 3000), \quad a(3000) = 4,500,751. \quad (9.1)$$

These are finite computational results, not an induction proving all n .

A separate Python standard-library implementation computes the triangle through row 1500 and agrees with the C++ term counts at all 1501 indices $0, \dots, 1500$. Further independent checks include the complete polynomials from the differential recurrence (2.2) through order 60, all 496 shifted-product coefficients (3.5) with $0 \leq r \leq m \leq 30$, and 11,865 instances of the prime-adic valuation formula. All 54 values displayed directly in the OEIS entry agree with the exact computation. The separately linked OEIS table through 500 was not used as an independent data source for this report.

The sixteen diagonal certificates were reconstructed and checked with a third, rational-polynomial computation. In addition to showing no unexpected zeros in a finite triangular region, these certificates prove there are no unexpected zeros on those fixed diagonals for any row index.

9.2 Algorithm and complexity

Only two preceding rows are required for (3.7). A schematic implementation is

```
previousprevious = [1]
previous = [0, 1]
count = 2
for m = 2, ..., N:
    current[0] = 0
    for r = 1, ..., m:
        current[r] = (r-m+1)*previous[r]
            + previous[r-1] + (m-1)*previousprevious[r-1]
    count += number of nonzero current[r] for 1 <= r <= m
    rotate the three rows
```

Indices beyond the end of a row mean zero. The supplied implementations handle these boundaries explicitly.

The computation uses $O(N^2)$ arithmetic operations and stores $O(N)$ integers. These are not bit-complexity claims: the integers grow with N . From the product formula,

$$|b(m, r)| \leq \prod_{h=0}^{m-1} (1 + |r - h|) \leq (m+1)^m,$$

so a coefficient has $O(m \log m)$ bits. The rolling-row implementation thus has an $O(N^2 \log N)$ -bit storage bound. A crude schoolbook arithmetic bound for the running time is $O(N^3 \log^2 N)$ bit operations.

9.3 Files and commands

The archive contains `article.tex`, the compiled `article.pdf`, a build script, both triangle verifiers, the rational certificate verifier, the complete count and zero-position tables through 3000, and the exact verification reports. From the extracted directory, run

```
python3 verify.py -max-n 1500 -data-dir data
python3 verify_diagonals.py -output-dir data -check-only
```

The C++ implementation requires a C++17 compiler and GMP development files:

```
c++ -O3 -std=c++17 verify_gmp.cpp -lgmpxx -lgmp -o verify_gmp
./verify_gmp 3000 data
```

The Python programs use only the standard library. The diagonal checker constructs the coefficients afresh rather than trusting the saved residue tables. Its proof obligations are elementary rational polynomial identities and finite modular evaluations; it does not require a general-purpose computer algebra system.

10 What remains to finish the requested proof

The derivation of the candidate rational function is complete, but the identification of that function with the derivative count is conditional. The exact outstanding statement is

$$[u^r] \prod_{h=0}^{m-1} (u + r - h) \neq 0$$

for all $1 \leq r \leq m$ outside the explicitly proved zero set $\mathcal{Z}_0$. The preceding results settle this for $r \leq 3$, for $m - r \leq 16$, and whenever $m - r + 1$ is prime, in addition to all rows through 3000 by exact computation. They do not cover every remaining pair.

The diagonal-polynomial formulation gives a particularly precise route to a complete proof. For every offset d , it would suffice to show that the residual polynomial R_d in (5.9) has no integer root $m > d$. Irreducibility would be stronger than necessary. A uniform family of modular nonvanishing certificates would also suffice, but the fact that such certificates work for $d \leq 16$ does not establish that they work for all d .

For the asymptotic conjecture alone, an exclusion of every additional zero is unnecessary: by (7.3), it suffices to bound their cumulative number by $o(n^2)$. The present root-counting argument proves quadratic order but not this sharper estimate.

Conclusion. The printed recurrence with range $n > 6$ is disproved, and its correct range for the proposed sequence is identified. The principal closed formula is supported by an exact structural reduction, several infinite nonvanishing theorems, and extensive reproducible verification. A complete all-order proof of that formula is *not* supplied: the missing converse in Conjecture 4.4 is left explicit rather than being replaced by an inference from numerical agreement.

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